E-commerce Customer Behavior Analysis and Business Insights

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Problem Statement

In the highly competitive e-commerce industry, understanding customer behavior is essential for improving sales and customer satisfaction. However, businesses often struggle to identify the key factors that influence customer spending, engagement, and retention. This project aims to analyze custome data to uncover patterns in purchasing behavior, evaluate the impact of discounts and membership types, and provide insights that help businesses make data-driven decisions.

objective

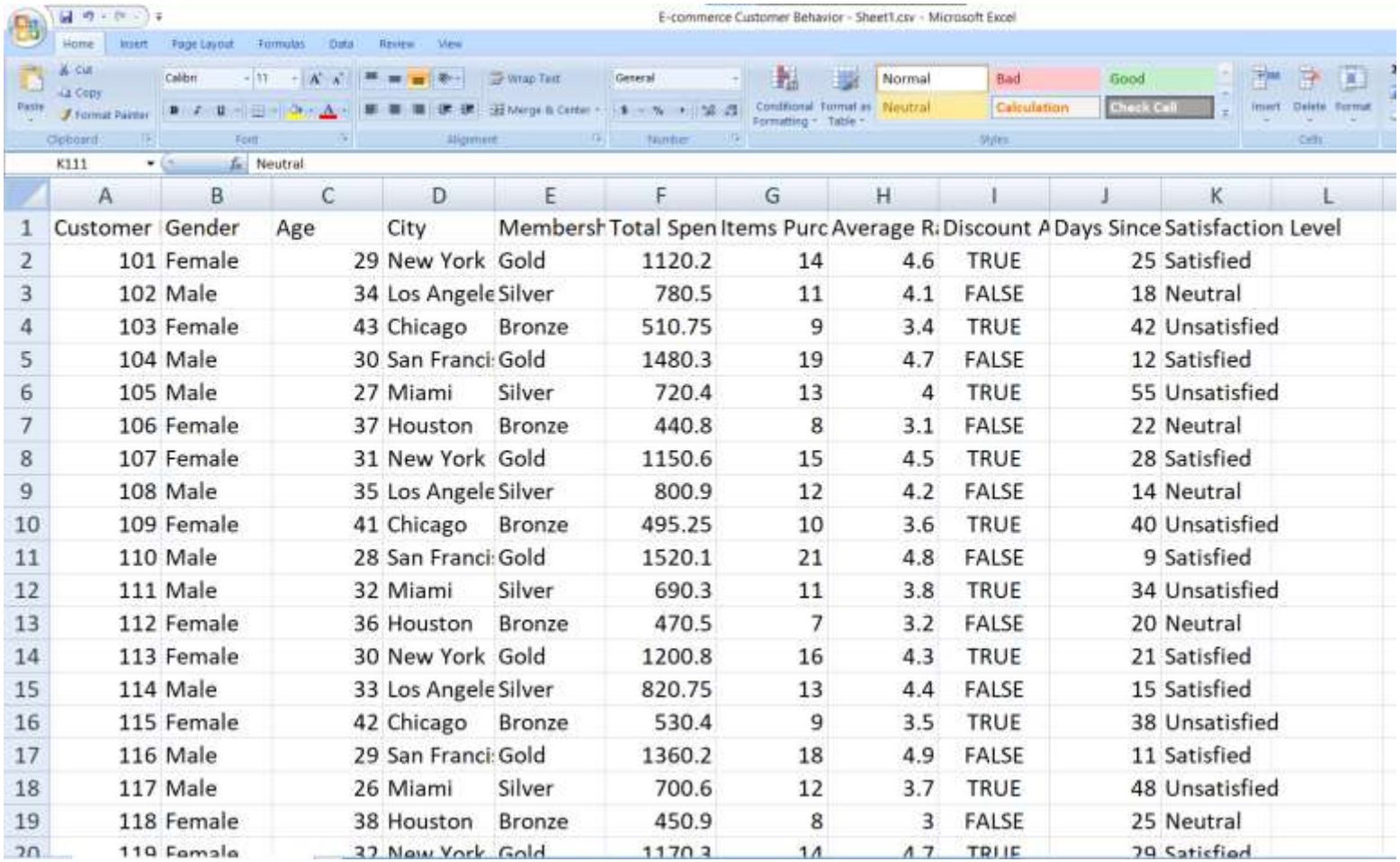
To analyze customer behavior in an e-commerce platform and identify key factors influencing customer spending, satisfaction, and retention.

- To examine the impact of discounts and membership types on customer spending
- To identify patterns in customer satisfaction and purchasing behavior
- To provide data-driven insights for improving business decisions

Dataset Overview

- The dataset contains detailed information about customer behavior in an e-commerce platform, including demographics, purchasing patterns, and satisfaction levels.
- **Customer ID** – Unique identifier for each customer
- **Age & Gender** – Demographic details **City** – Location of the customer **Membership Type** – Gold, Silver, Bronze
- **Total Spend** – Total amount spent by the customer
- **Items Purchased** – Number of items bought
- **Discount Applied** – Whether discount was used
- **Average Rating** – Customer rating (0–5)
- **Days Since Last Purchase** – Customer activity level **Satisfaction Level** – Satisfied, Neutral, Unsatisfied

Data set view



	A	B	C	D	E	F	G	H	I	J	K	L
1	Customer	Gender	Age	City	Membersh	Total Spen	Items Purc	Average R	Discount A	Days Since	Satisfaction Level	
2	101	Female	29	New York	Gold	1120.2	14	4.6	TRUE	25	Satisfied	
3	102	Male	34	Los Angele	Silver	780.5	11	4.1	FALSE	18	Neutral	
4	103	Female	43	Chicago	Bronze	510.75	9	3.4	TRUE	42	Unsatisfied	
5	104	Male	30	San Franci	Gold	1480.3	19	4.7	FALSE	12	Satisfied	
6	105	Male	27	Miami	Silver	720.4	13	4	TRUE	55	Unsatisfied	
7	106	Female	37	Houston	Bronze	440.8	8	3.1	FALSE	22	Neutral	
8	107	Female	31	New York	Gold	1150.6	15	4.5	TRUE	28	Satisfied	
9	108	Male	35	Los Angele	Silver	800.9	12	4.2	FALSE	14	Neutral	
10	109	Female	41	Chicago	Bronze	495.25	10	3.6	TRUE	40	Unsatisfied	
11	110	Male	28	San Franci	Gold	1520.1	21	4.8	FALSE	9	Satisfied	
12	111	Male	32	Miami	Silver	690.3	11	3.8	TRUE	34	Unsatisfied	
13	112	Female	36	Houston	Bronze	470.5	7	3.2	FALSE	20	Neutral	
14	113	Female	30	New York	Gold	1200.8	16	4.3	TRUE	21	Satisfied	
15	114	Male	33	Los Angele	Silver	820.75	13	4.4	FALSE	15	Satisfied	
16	115	Female	42	Chicago	Bronze	530.4	9	3.5	TRUE	38	Unsatisfied	
17	116	Male	29	San Franci	Gold	1360.2	18	4.9	FALSE	11	Satisfied	
18	117	Male	26	Miami	Silver	700.6	12	3.7	TRUE	48	Unsatisfied	
19	118	Female	38	Houston	Bronze	450.9	8	3	FALSE	25	Neutral	
20	119	Female	32	New York	Gold	1170.3	14	4.7	TRUE	29	Satisfied	

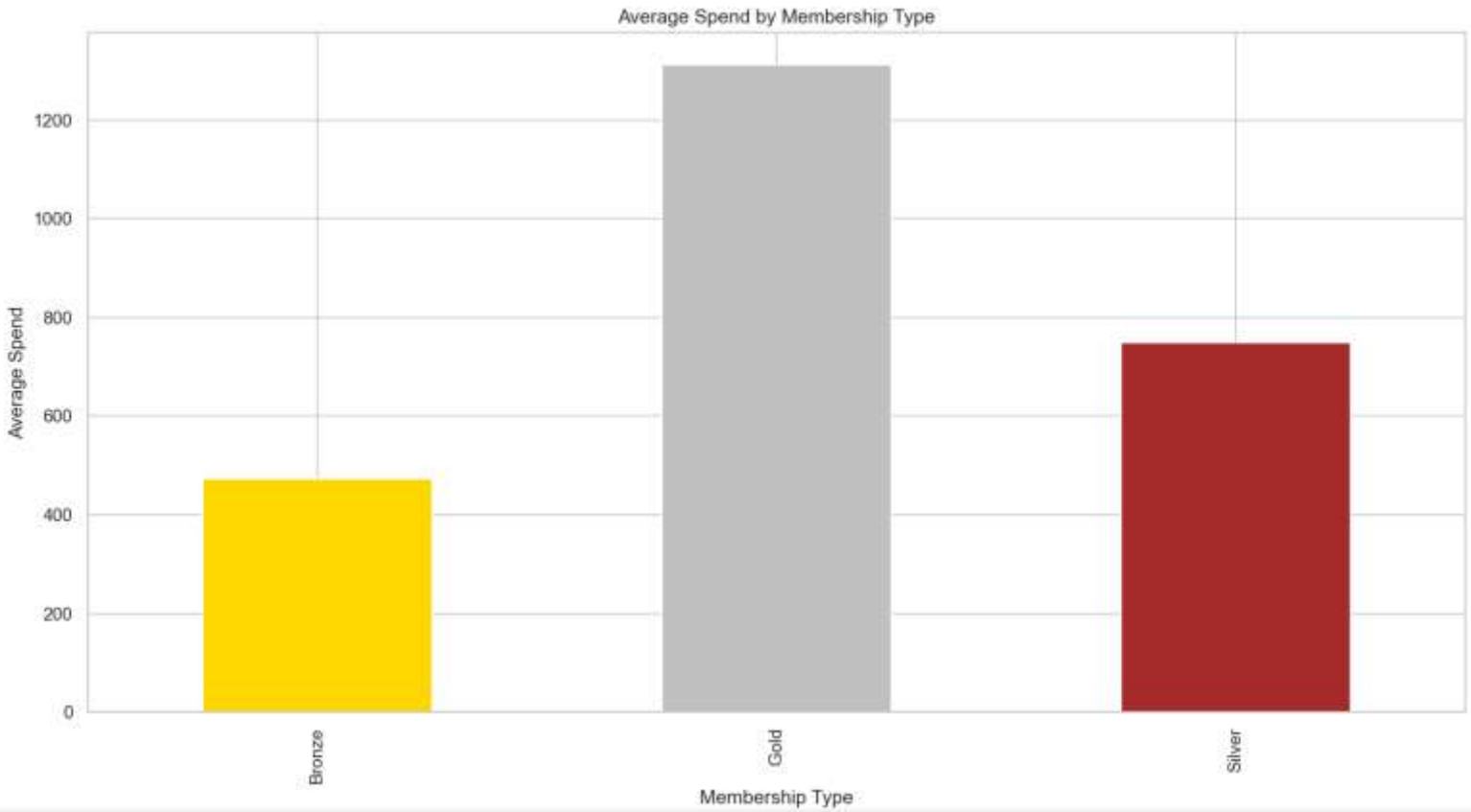
Tools & Technologies

- Python
- Pandas
- Matplotlib
- Seaborn
- Statistical Testing (T-test)

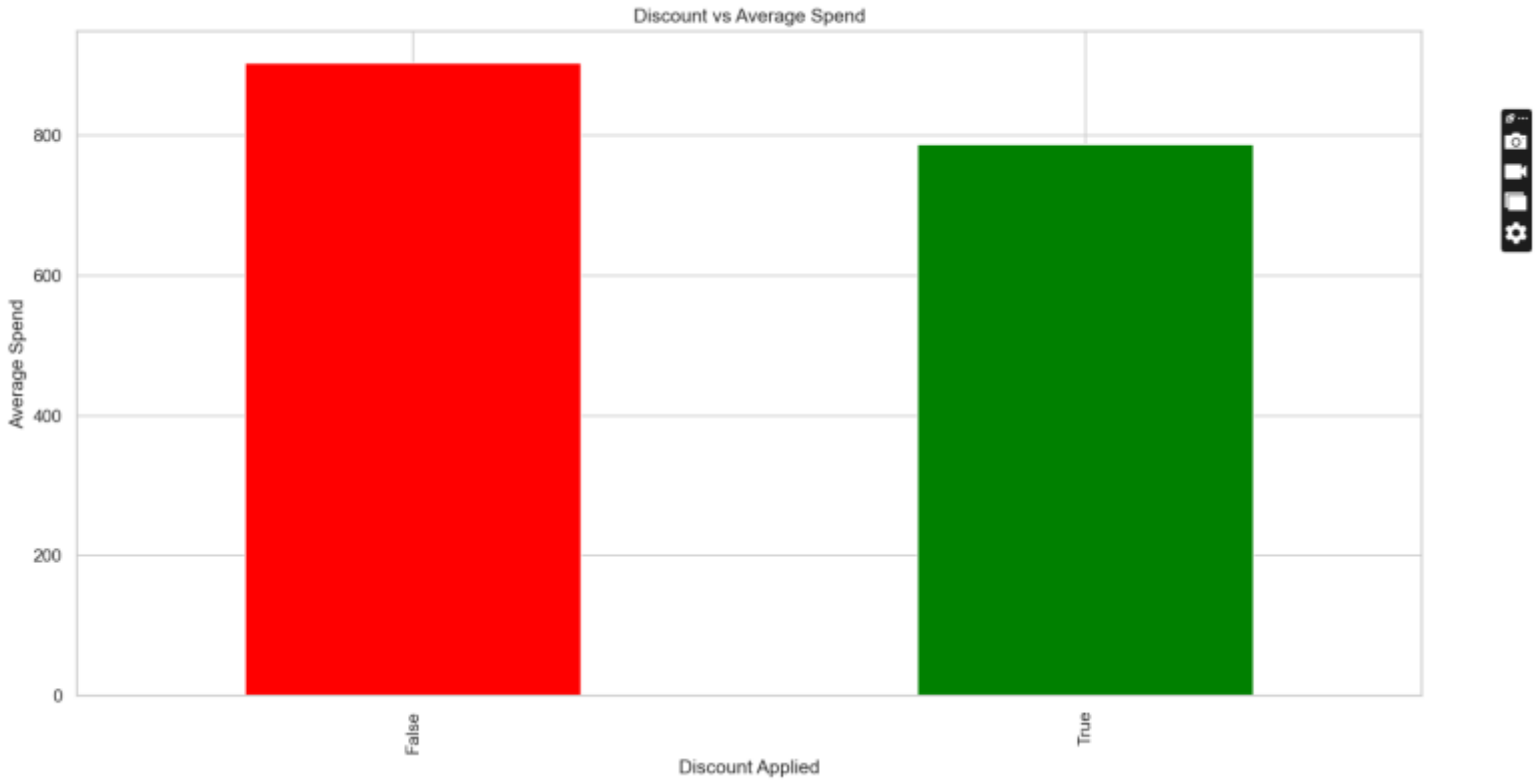
Data Analysis Approach

- **Steps Followed:**
- **Data Cleaning**
  Removed inconsistencies, handled missing values, and formatted column names
- **Exploratory Data Analysis (EDA)**
  Analyzed patterns in customer behavior using statistical summaries
- **Data Visualization**
  Created charts (bar plots, histograms, boxplots) to understand trends
- **Feature Analysis**
  Studied relationships between variables like:
  - Membership Type vs Spending
  - Discount vs Spending
  - Satisfaction vs Spending
- **Hypothesis Testing**
  Performed a T-test to evaluate the impact of discounts on customer spending

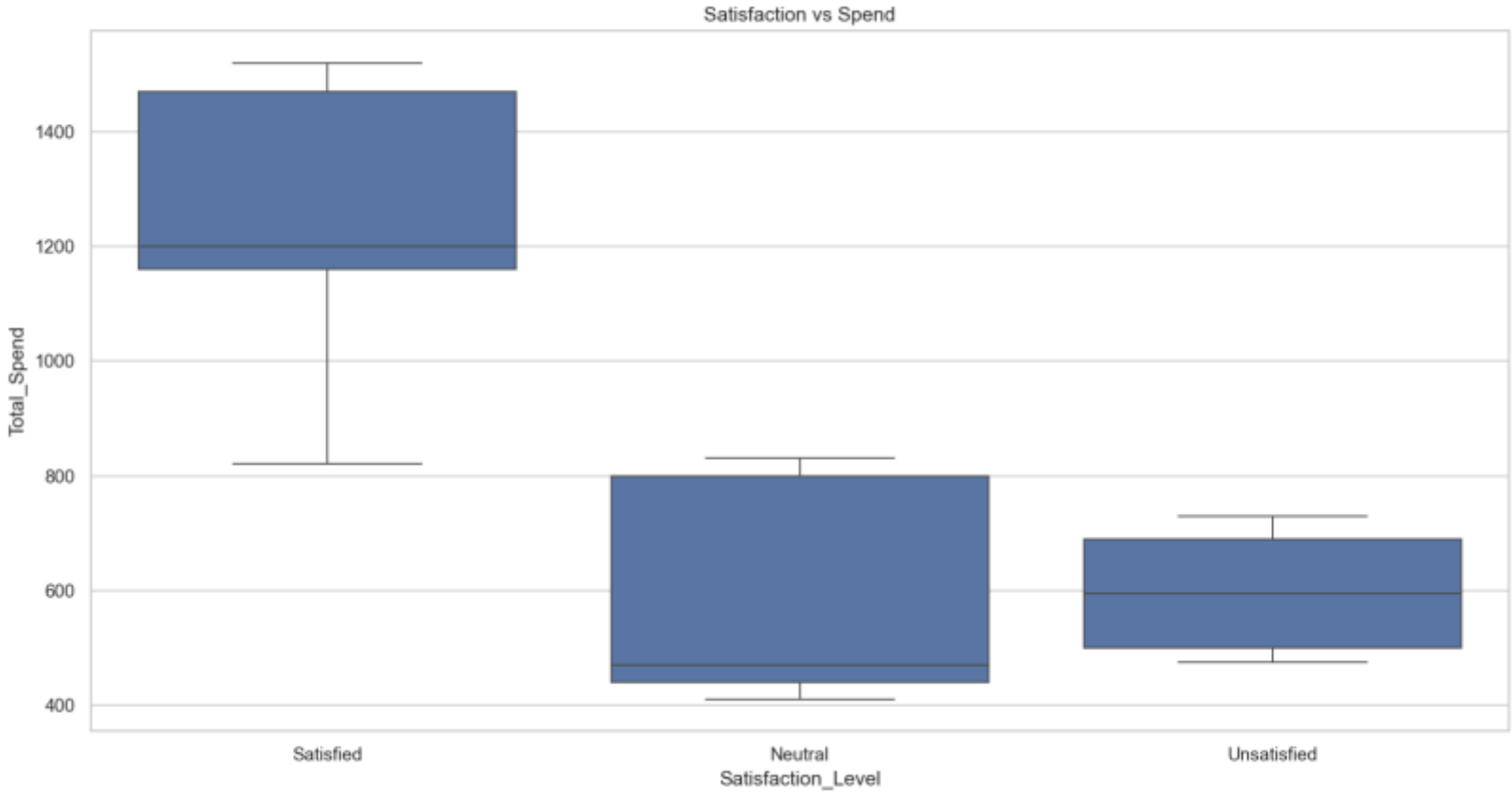
Key Insight 1



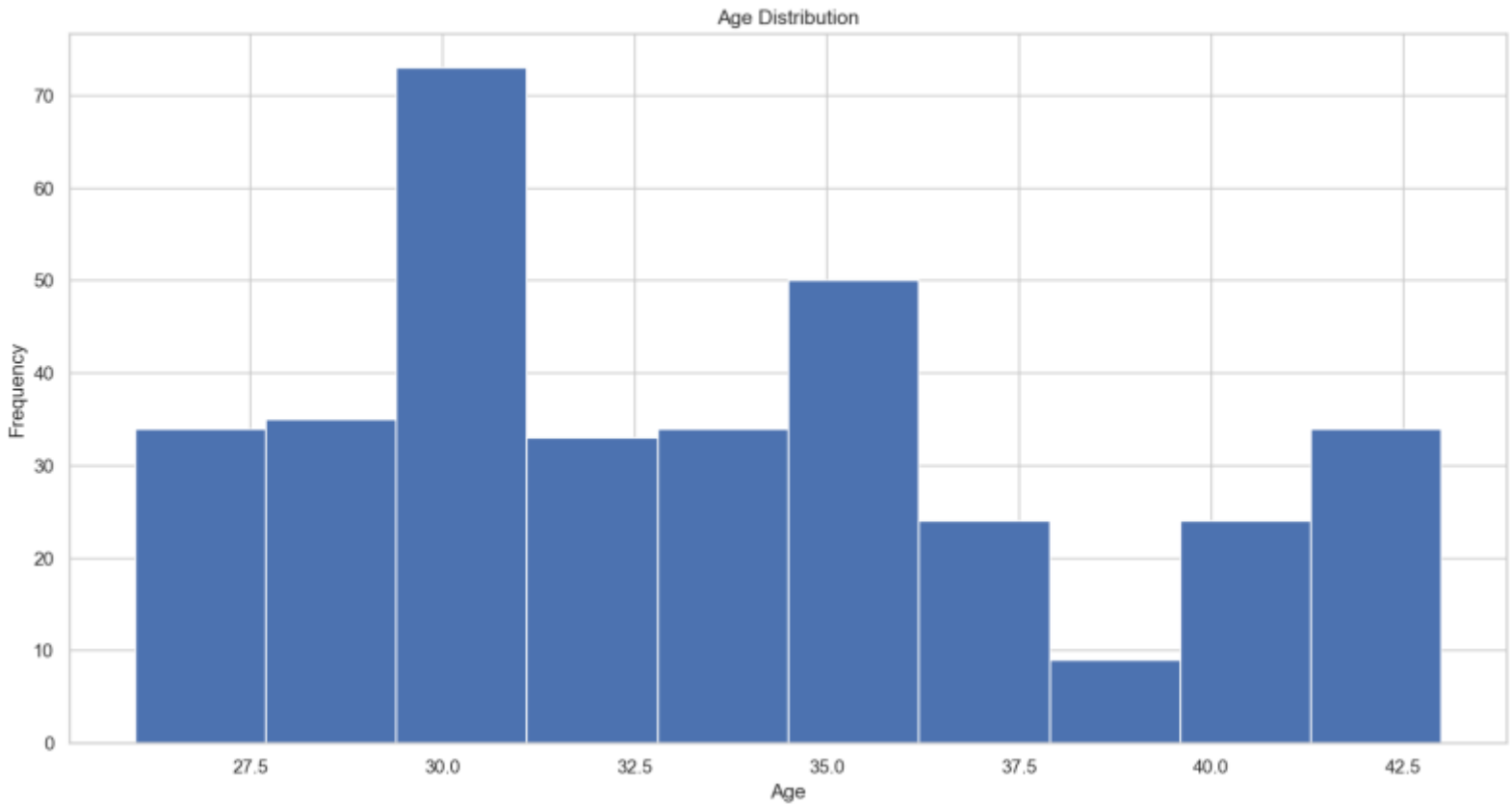
Key Insight 2



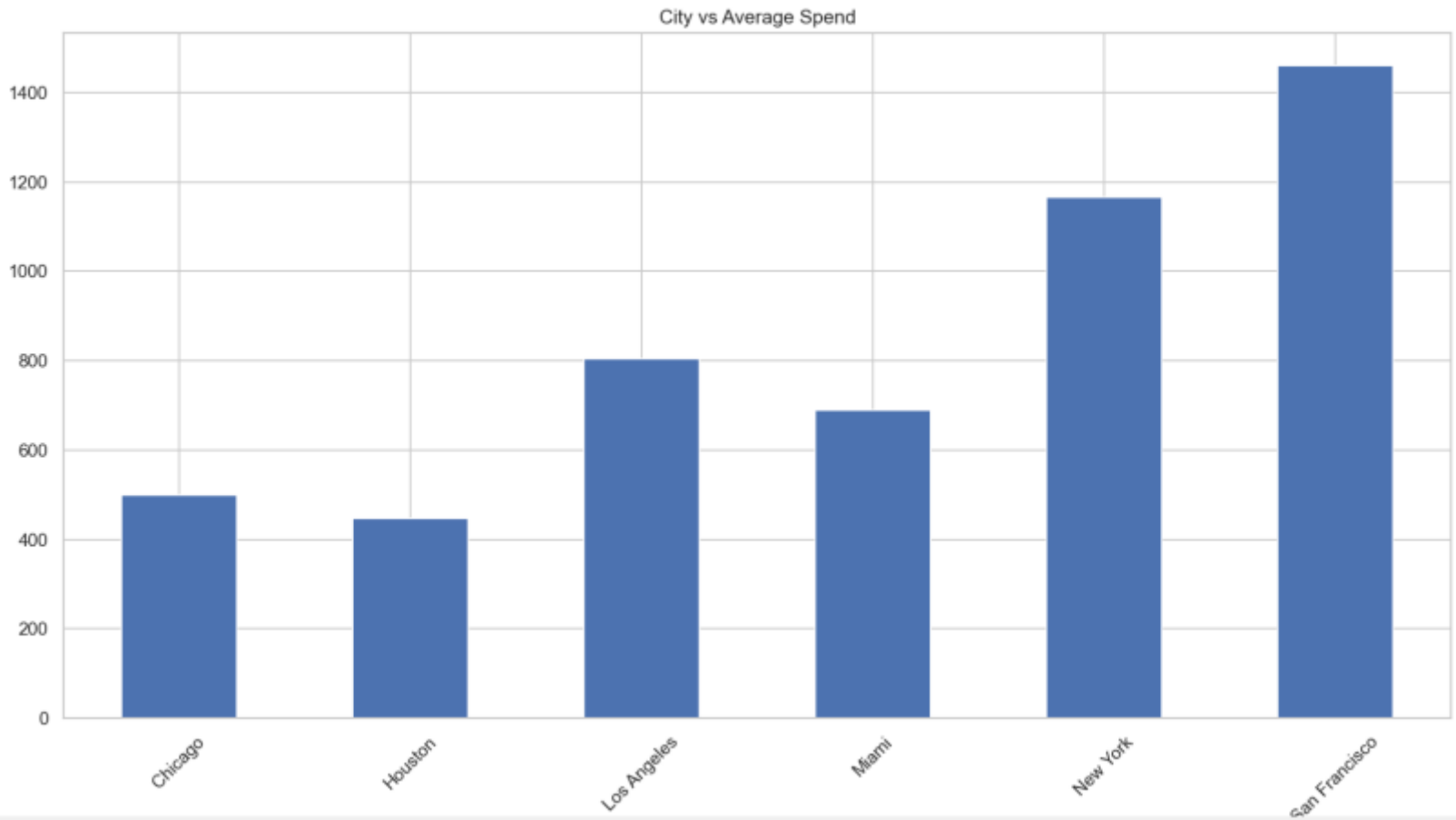
Key Insight 3



Key Insight 4



Key Insight 5



Hypothesis Testing

- **Objective:**
- To determine whether discounts have a significant impact on customer spending.
- ◆ **Hypotheses:**
- **H_0 (Null Hypothesis):**
  Discount has no effect on customer spending
- **H_1 (Alternative Hypothesis):**
  Discount significantly affects customer spending
- ◆ **Statistical Test Used:**
- **T-test (Independent Sample T-test)**
- Compared average spending of:
  - Customers with discount
  - Customers without discount
- ◆ **Results:**
- **T-statistic:** *(add your value)*
- **P-value:** *(add your value)*
- ◆ **Conclusion:**
- 👉 If **$p < 0.05$**:
- Reject H_0 — Discounts have a significant impact on customer spending
- 👉 If **$p > 0.05$**:
- Fail to reject H_0 — Discounts do not have a significant impact

Results and Discussion

- ◆ **Key Results:**
- Customer spending varies significantly across different membership types
- Discounts influence customer purchasing behavior
- Satisfied customers tend to spend more
- A portion of customers shows inactivity, indicating retention challenges
- ◆ **Statistical Result:**
- 👉 Based on hypothesis testing:
- **P-value = (add your value)**
- ✓ If **p < 0.05**:
- Discounts have a statistically significant impact on customer spending
- ✓ If **p > 0.05**:
- Discounts do not have a statistically significant impact

Business Recommendations

- **1. Implement Targeted Discount Strategies**
- Offer discounts to specific customer segments
- Use data to identify customers more likely to respond
- ◆ **2. Focus on High-Value Customers**
- Prioritize **Gold membership** customers
- Provide exclusive offers and loyalty benefits
- ◆ **3. Improve Customer Satisfaction**
- Address issues faced by unsatisfied customers
- Enhance product quality and customer service
- ◆ **4. Re-engage Inactive Customers**
- Target customers with high **Days Since Last Purchase**
- Use personalized emails, offers, and reminders
- ◆ **5. Optimize Marketing Based on Insights**
- Use city and demographic data for targeted campaigns
- Focus on high-performing regions

Conclusion

This project analyzed customer behavior in an e-commerce platform to identify key factors influencing spending and satisfaction. The results show that membership type, discounts, and customer satisfaction play a significant role in purchasing behavior. Statistical analysis further validated the impact of discounts on spending. These insights can help businesses improve customer engagement, increase revenue, and make data-driven decisions.